

Mario Makhlouta

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OBJECTIVE

Computer & Software Engineering student with a strong foundation in programming, embedded systems, and cybersecurity. Hands-on experience through internships in cybersecurity and robotics, plus academic and personal projects in AI, FPGA design, and full-stack development. Seeking opportunities to apply my skills, contribute to impactful projects, and grow further in software engineering and any related fields.

EDUCATION

Bachelor of Engineering (BE) in Computer Engineering — University of Balamand (UOB) 2026 – 2027

Faculty of Engineering. One-year BE track following the BS in Computer Engineering.

Software Engineering Student — 42 Beirut

2024 – August 2026

Peer-to-peer learning, project-based curriculum in C/C++. Completed through Milestone 4. Projects: push_swap, FdF, minishell, cub3D, C++ modules.

Bachelor of Science (BS) in Computer Engineering — University of Balamand (UOB) 2022 – May 2026

Relevant courses: Data Structures, AI for Engineers, Operating Systems, Telecommunications Lab, Microprocessors, Programmable Logic Controllers (PLC), Cybersecurity, Computer Embedded Systems, Computer Architecture, Electronics.

EXPERIENCE

Robotics Intern & Trainee — InMind Academy / InMind Company

July – August 2026

- Completed a two-month combined training and internship program in the robotics track.
- Academy phase covered robotics fundamentals: kinematics, sensor integration, actuator control, and autonomous navigation principles.
- Company internship phase applied knowledge to real hardware: programming embedded controllers, wiring sensor arrays, and building autonomous motion routines.
- Earned dual certification: InMind Academy training certificate and InMind Company internship certificate.

Cybersecurity Intern — Prodigy InfoTech (Remote)

April 2026

- Completed 5 hands-on cybersecurity tasks across a one-month remote internship.
- Task 1: Caesar cipher tool — encryption, decryption, and brute-force analysis of all 26 shifts.
- Task 2: Pixel-level image encryption tool using XOR keying and RGB channel swapping.
- Task 3: Password strength analyser scoring across five complexity criteria with improvement guidance.
- Task 4: Consent-gated educational keylogger with timestamped session logging.
- Task 5: Network packet sniffer capturing live traffic — IP addresses, protocols, ports, and payload previews.
- Each task delivered as a standalone Python tool with documentation and ethical-use safeguards.

TECHNICAL SKILLS

Programming	C, C++, Python, Java, SQL, Bash, Assembly (RISC-V, PIC18), SystemVerilog, PLC Ladder Logic, MATLAB
AI & Data	AI for Engineers fundamentals, Data Structures & Algorithms
Systems & OS	Operating Systems concepts, Unix/Linux, Memory management, Processes & threads
Embedded / FPGA	Quartus Prime, MPLAB, Multisim, Proteus, Arduino, PIC18, DE1-SoC, NIOS-V, Pipelined Processor Design, CPU Pipeline Hazards
Cybersecurity	Penetration testing fundamentals, Privilege escalation (Linux/Unix), Network security basics, Ethical hacking practice

Telecommunications	SSB-AM simulation, Telecom Lab measurements, Signal analysis
Web & Software	Object-Oriented Programming, Backend (Python), Full-stack basics, Git, Visual Studio
Databases	SQL, MariaDB, MS Access
Tools	Git/GitHub, Simulink, AutoCAD

LANGUAGES

Arabic (native) • English (intermediate) • French (intermediate)

CERTIFICATES

Cybersecurity Internship Completion — Prodigy InfoTech	2026
Robotics Internship & Training — InMind Company / InMind Academy	2026
OCB-051 Cybersecurity Bootcamp — Semicolon Academy	2025
IEEE Membership Certificate	2024
GDSC Board Membership Certificate — Google	2024
First Aid & CPR — Lebanese Red Cross	June 2024
42 Beirut Piscine Attendance Certificate	April 2024
MATLAB Onramp — MathWorks	March 2023
Robotics Training Program — GIL	2022
Full Stack Development Track — 1 million Arab Coders Initiative	November 2019

EXTRACURRICULAR ACTIVITIES

- **Member — Institute of Electrical and Electronics Engineers (IEEE), UOB**
- **Arduino-Robotics Training Program — Generation of Innovation Leaders (GIL)** — final project: smart box with sensor-controlled access.